ADHAM ALREFAI

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SUMMARY

Biomedical Engineer with a strong background in machine learning and medical device maintenance. Proven track record in developing high-accuracy clinical predictive models and ensuring the operational excellence of laboratory instrumentation. Passionate about leveraging AI and full-stack development to advance clinical decision support and healthcare innovation.

EDUCATION

Bachelor of Science in Biomedical Engineering | *2020 – 2025*
Modern University for Technology & Information (MTI), Cairo, Egypt

- Graduation Project: Microscope System & Computational Model for Leukemia Detection — combined hardware design with a deep learning classifier, achieving 99.99% test accuracy and 95% accuracy on real hospital blood cell images
- Focus areas: bio-electronics, medical device systems, and applied machine learning for clinical decision support

EXPERIENCE

Technical Lead, Machine Learning | *Feb – Apr 2026*
AI CoLab — MedStar-Georgetown Collaborative Center for Artificial Intelligence in Healthcare Research and Education

- Led the ED Overcrowding Prediction project, developing machine learning models to forecast emergency department overcrowding transitions for the next clinical shift using the MIMIC-IV-ED dataset
- Engineered advanced clinical features, including qSOFA score computation and pain-score standardization, to strengthen model predictive signal

Biomedical Engineer | *Oct – Dec 2025*
Biomedical Maintenance Enterprise (BME)

- Ensured adherence to safety and quality protocols through rigorous maintenance and calibration of laboratory instruments and hematology analyzers.
- Resolved issues with intricate medical equipment and oversaw preventative maintenance programs in compliance with industry regulations.
- Boosted laboratory productivity by minimizing device downtime through strategic maintenance and proficient diagnostic interventions.
- Defined the predictive target by resampling raw admissions data into 8-hour clinical shifts aligned with hospital staffing cycles
- Implemented SHAP-based model interpretability to support clinical transparency and build stakeholder trust in model outputs

Biomedical Engineering Trainee | *Jul – Aug 2025*
The Children's Cancer Hospital 57357, Cairo, Egypt

- Observed core biomedical engineering operations within a high-acuity hospital environment
- Studied hospital medical equipment planning, full device lifecycle management, and Quality Assurance (QA) procedures
- Gained hands-on exposure to device installation, preventive maintenance scheduling, and calibration protocols

POSITIONS OF RESPONSIBILITY

Technical Content Writer, Medium (@adham_ayman)*2025 – Present*

- Author tutorials on explainable AI concepts, including SHAP analysis, model calibration, and evaluation metrics
- Simplify advanced machine learning concepts for students and practitioners to broaden community access and learning

TECHNICAL SKILLS

Languages: Arabic: Mother tongue, English: Excellent
Programming Languages: Python, C/C++, JavaScript, HTML, CSS
Libraries / Frameworks: NumPy, Pandas, Scikit-learn, TensorFlow, Keras, LangChain, LangGraph, FastAPI, React.js, Node.js
Tools: Git, GitHub, Docker, AWS, MongoDB, Weights & Biases
Areas of Interest: Biomedical Engineering, AI in Healthcare, LLM Engineering, RAG Systems,Full-Stack Development